Robust Bridge Sampling:

Diagnostics for Influence, Conditioning, and Proposal Variance

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Table of Contents

Table of Contents

Motivation

- We often need the marginal likelihood (also known as model evidence)

$$\mathcal{Z} \;=\; p(y \mid \mathcal{M}) \;=\; \int p(y \mid \theta, \mathcal{M})\, p(\theta \mid \mathcal{M})\, d\theta.$$

- Bridge sampling is a widely used approach to estimate $\mathcal{Z}$ and Bayes factors.
- Practical pain point: a few draws can dominate, fixed-point map can be poorly conditioned, and fitting the proposal adds variance.
- **Goal:** fast diagnostics with clear rules and calibrated thresholds.

Setup And Bridge Identity

Let $f(\theta) := p(y \mid \theta, \mathcal{M})\, p(\theta \mid \mathcal{M})$ be the unnormalized posterior, and let $g(\theta \mid \vartheta)$ be a proposal.

Bridge Identity (For Any Measurable h)

$$\mathcal{Z} = \frac{\mathbb{E}_g\big[f(\theta)\, h(\theta)\big]}{\mathbb{E}_{p(\theta \mid y, \mathcal{M})}\big[g(\theta \mid \vartheta)\, h(\theta)\big]}. \tag{1}$$

With draws $\{\tilde{\theta}_i\}_{i=1}^{S_2} \sim g$ and $\{\theta_j^*\}_{j=1}^{S_1} \sim p(\theta \mid y, \mathcal{M})$:

$$\widehat{\mathcal{Z}} = \frac{\bar{N}}{\bar{D}},^{1} \qquad \bar{N} := \frac{1}{S_2} \sum_{i=1}^{S_2} f(\tilde{\theta}_i)\, h(\tilde{\theta}_i), \quad \bar{D} := \frac{1}{S_1} \sum_{j=1}^{S_1} g(\theta_j^* \mid \vartheta)\, h(\theta_j^*). \tag{2}$$

[1] The normalizing constant of h cancels.

Variance-Reducing Bridge Function

Let $s_1 := S_1/(S_1 + S_2)$ and $s_2 := S_2/(S_1 + S_2)$.

$$h^\star(\theta) \;\propto\; \frac{1}{s_1 f(\theta) + s_2\, \mathcal{Z}\, g(\theta \mid \vartheta)} \tag{3}$$

Claims:

- $h^\star$ minimizes the delta-method asymptotic variance of the bridge estimator $\widehat{\mathcal{Z}} = \bar{N}/\bar{D}$ among all measurable h with finite second moments, when using S_1 posterior and S_2 proposal draws (independent).
- The transformed summands are *bounded*: $\bar{N} =: u_2(\theta; \mathcal{Z}) \in [0, s_1^{-1}]$, $\bar{D} =: v_1(\theta; \mathcal{Z}) \in \left(0, (s_2\mathcal{Z})^{-1}\right]$, which damps tail-driven variance.

Delta-Variance Objective

Let

$$A(h) := \mathbb{E}_g[fh] = \int f(\theta)h(\theta)\,g(\theta)\,d\theta,$$

$$B(h) := \mathbb{E}_{p(\theta|y,\mathcal{M})}[gh] = \int g(\theta)h(\theta)\,p(\theta \mid y,\mathcal{M})(\theta)\,d\theta.$$

Note that $A(h) = \mathcal{Z}\,B(h)$ for *any* h since $p(\theta \mid y,\mathcal{M}) = f/\mathcal{Z}$. The delta method gives

$$\begin{aligned}
\mathrm{Var}(\widehat{\mathcal{Z}}) &\approx \frac{\mathrm{Var}_g(fh)}{S_2\,B(h)^2} + \frac{A(h)^2}{S_1\,B(h)^4}\,\mathrm{Var}_{p(\theta|y,\mathcal{M})}(gh) \\
&= \frac{1}{B(h)^2}\left[\frac{\mathbb{E}_g\big[(fh)^2\big] - A(h)^2}{S_2} + \frac{\mathcal{Z}^2\left(\mathbb{E}_{p(\theta|y,\mathcal{M})}[(gh)^2] - B(h)^2\right)}{S_1}\right].
\end{aligned}$$

Delta-Variance Objective

Because scaling $h \mapsto ch$ cancels in $\widehat{\mathcal{Z}}$, we can impose $B(h) = 1$ and minimize only:

$$\min_{h:\, B(h)=1} \quad \frac{1}{S_2}\mathbb{E}_g\big[(fh)^2\big] + \frac{\mathcal{Z}^2}{S_1}\mathbb{E}_{p(\theta|y,\mathcal{M})}\big[(gh)^2\big].$$

Writing these as integrals and using $p(\theta \mid y, \mathcal{M}) = f/\mathcal{Z}$,

$$\mathbb{E}_g\big[(fh)^2\big] = \int f^2 h^2\, g\, d\theta,$$

$$\mathbb{E}_{p(\theta|y,\mathcal{M})}\big[(gh)^2\big] = \frac{1}{\mathcal{Z}}\int g^2 h^2\, f\, d\theta,$$

$$B(h) = \frac{1}{\mathcal{Z}}\int f\, g\, h\, d\theta = 1.$$

Solve The Constrained Problem

Thus we minimize the quadratic functional

$$\mathcal{J}(h) = \int \Big[\underbrace{\tfrac{1}{S_2} f^2 g}_{\text{proposal term}} + \underbrace{\tfrac{\mathcal{Z}}{S_1} g^2 f}_{\text{posterior term}} \Big] h(\theta)^2 \, d\theta \quad \text{subject to} \quad \int f\, g\, h\, d\theta = \mathcal{Z}.$$

Introduce a Lagrange multiplier λ for the linear constraint and differentiate:

$$\mathcal{L}(h) = \int \Big[\tfrac{1}{S_2} f^2 g + \tfrac{\mathcal{Z}}{S_1} g^2 f\Big] h^2 \, d\theta - \lambda\Big(\int f\, g\, h\, d\theta - \mathcal{Z}\Big).$$

First-order condition (pointwise in θ):

$$2\Big(\tfrac{1}{S_2} f^2 g + \tfrac{\mathcal{Z}}{S_1} g^2 f\Big)\, h(\theta) - \lambda f(\theta) g(\theta) = 0 \quad \Longrightarrow \quad h(\theta) \propto \frac{f(\theta) g(\theta)}{\frac{1}{S_2} f(\theta)^2 g(\theta) + \frac{\mathcal{Z}}{S_1} g(\theta)^2 f(\theta)}.$$

Solve The Constrained Problem

And thus, after cancelling a common fg factor:

$$\boxed{h(\theta) \;\propto\; \frac{1}{f(\theta)/S_2 + \mathcal{Z}\, g(\theta)/S_1} \;=\; \frac{1}{\frac{1}{s_2(S_1+S_2)} f + \frac{\mathcal{Z}}{s_1(S_1+S_2)} g} \;\propto\; \frac{1}{s_1 f(\theta) + s_2\, \mathcal{Z}\, g(\theta \mid \vartheta)}}\,.$$

Intuition:

- The denominator $s_1 f + s_2 \mathcal{Z} g$ is large in regions where either density is large, so the transformed terms u_2, v_1 *saturate* instead of exploding, this is a harmonic-type damping of extreme importance ratios.
- The weights s_1, s_2 trade off where variance is coming from: when $S_1 \gg S_2$, the larger sample contributes less to the variance budget, and the optimal h downweights its term accordingly.

Caveats

- The derivation optimizes the *delta* variance and assumes finite second moments and sufficient support overlap; with poor overlap no choice of h can rescue the estimator.
- $\mathcal{Z}$ is unknown; in practice we plug in $\widehat{\mathcal{Z}}$ via the fixed-point map. This adds small extra randomness, quantified by our iteration-conditioning and jackknife diagnostics.
- Use *ESS* in s_1, s_2 when chains are dependent; otherwise the weighting is mis-calibrated.

Fixed-Point Map

Since h depends on $\mathcal{Z}$, compute $\widehat{\mathcal{Z}}$ via fixed-point iteration. Define importance ratios

$$\ell_{1,j} := \frac{f(\theta_j^*)}{g(\theta_j^* \mid \vartheta)}, \tag{4}$$

$$\ell_{2,i} := \frac{f(\tilde{\theta}_i)}{g(\tilde{\theta}_i \mid \vartheta)}. \tag{5}$$

Then the update is

$$\widehat{\mathcal{Z}}^{(t+1)} = \frac{\frac{1}{S_2}\sum_{i=1}^{S_2} \frac{\ell_{2,i}}{s_1 \ell_{2,i} + s_2 \widehat{\mathcal{Z}}^{(t)}}}{\frac{1}{S_1}\sum_{j=1}^{S_1} \frac{1}{s_1 \ell_{1,j} + s_2 \widehat{\mathcal{Z}}^{(t)}}}. \tag{6}$$

Conditional MCSE And What It Misses

Conditioning on the realized h and $g(\cdot \mid \vartheta)$,

$$\mathrm{Var}_{\mathrm{MC}}\left(\widehat{Z} \mid h, g\right) \approx \frac{\mathrm{Var}(\bar{N})}{\bar{D}^2} + \frac{\bar{N}^2}{\bar{D}^4}\,\mathrm{Var}(\bar{D}), \tag{7}$$

with standard ESS corrections for autocorrelation.

Gaps:

- Ignores extra variability from (i) the z-dependent transformation, (ii) fixed-point solution itself, (iii) fitting ϑ.
- Tail checks on raw ratios are masked by bounded summands in the map.

This motivates three diagnostics.

Table of Contents

Core Idea

Write the fixed-point as the root of a sample moment:

$$\Psi_S(z) := \frac{1}{S_2}\sum_{i=1}^{S_2} \underbrace{\frac{\ell_{2,i}}{s_1\ell_{2,i} + s_2 z}}_{u_2(\tilde{\theta}_i, z)} - z\,\frac{1}{S_1}\sum_{j=1}^{S_1} \underbrace{\frac{1}{s_1\ell_{1,j} + s_2 z}}_{v_1(\theta_j^*, z)} = 0. \tag{8}$$

Delete-a-group jackknife:

- Partition posterior and proposal draws into G contiguous blocks with sizes at least their integrated autocorrelation times (IACTs).
- For block g, *delete both samples block*, recompute $\widehat{\mathcal{Z}}_{(-g)}$.

Diagnostics

$$\widehat{\mathsf{Var}}_{\text{jack}}(\widehat{\mathcal{Z}}) := \frac{G-1}{G}\sum_{g=1}^{G}\left(\widehat{\mathcal{Z}}_{(-g)} - \overline{\mathcal{Z}}_{(-)}\right)^2,$$

$$\mathcal{V}_{\text{jack}} := \frac{\widehat{\mathsf{Var}}_{\text{jack}}(\widehat{\mathcal{Z}})}{\mathsf{Var}_{\text{MC}}(\widehat{\mathcal{Z}})}.$$

$$\text{MaxInfluence} := \max_{g} \frac{\left|\widehat{\mathcal{Z}}_{(-g)} - \widehat{\mathcal{Z}}\right|}{\sqrt{\mathsf{Var}_{\text{MC}}(\widehat{\mathcal{Z}})}}.$$

Interpretation

- If $\mathcal{V}_{\text{jack}} \gg 1$ or $\mathrm{MaxInfluence}$ is large, a small subset of draws exerts disproportionate leverage on the fixed point, a pre-asymptotic fragility *not* seen by the conditional delta MCSE.
- Asymptotically (under mild overlap, mixing, unique root):
  - $\widehat{\mathsf{Var}}_{\text{jack}}(\widehat{\mathcal{Z}})$ is consistent for the *unconditional* MC variance (adds the map/fit noise).
  - Deletion effects approximate sums of influence functions over the deleted blocks.

Thresholds

Let $\mathrm{Var}_{\mathrm{MC}}(\widehat{\mathcal{Z}}) = v_1 + v_2$ be posterior/proposal contributions.

Family-Wise Bound For $\mathrm{MaxInfluence}$

For balanced blocks ($m_k \approx S_k/G$), define shares $w_1 := \frac{v_1}{v_1+v_2}$, and $w_2 := 1 - w_1$, then

$$\Pr\big(\mathrm{MaxInfluence} > t_\alpha\big) \;\lesssim\; \alpha, \qquad t_\alpha = \frac{\Phi^{-1}(1-\alpha/(2G))}{\sqrt{G}}.$$

Variance-Inflation Test For $\mathcal{V}_{\mathrm{jack}}$

Under the well-behaved "null" ($\mathrm{Var}_{\mathrm{uncond}} = \mathrm{Var}_{\mathrm{MC}}$),

$$\frac{(G-1)\,\widehat{\mathrm{Var}}_{\mathrm{jack}}(\widehat{\mathcal{Z}})}{\mathrm{Var}_{\mathrm{MC}}(\widehat{\mathcal{Z}})} \;\dot\sim\; \chi^2_{G-1} \quad \Rightarrow \quad \mathcal{V}_{\mathrm{jack}} > \frac{\chi^2_{G-1}(1-\alpha)}{G-1} \text{ flags underestimation.}$$

Practitioner View

- Report $(\mathcal{V}_{\text{jack}},\ \text{MaxInfluence})$ *with* thresholds at $\alpha \in \{0.10, 0.05, 0.01\}$.
- If $\mathcal{V}_{\text{jack}} > \mathcal{V}_{\text{thr}}(0.05)$ or $\text{MaxInfluence} > t_{0.05}$:
  - Conclude the conditional MCSE understates uncertainty.
  - Remedies: enlarge/retarget proposal; increase S_1, S_2; thin to reduce dependence; re-fit ϑ with more draws.

Table of Contents

Local Conditioning Of The Fixed-Point Map

Let $T(z) = n(z)/d(z)$ with

$$n'(z) = -\frac{s_2}{S_2}\sum_{i=1}^{S_2}\frac{\ell_{2,i}}{(s_1\ell_{2,i} + s_2 z)^2}, \qquad d'(z) = -\frac{s_2}{S_1}\sum_{j=1}^{S_1}\frac{1}{(s_1\ell_{1,j} + s_2 z)^2}.$$

$$T'(z) = \frac{n'(z)d(z) - n(z)d'(z)}{d(z)^2}, \qquad C := \left|T'(\widehat{z})\right|.$$

A Posteriori Iteration-Error Proxy

Let $r_t := z^{(t+1)} - z^{(t)}$ at the last step. Then

$$\delta_{\mathrm{iter}} := \frac{|r_t|}{1 - C} \quad \text{(upper-bounds the distance to the } \textit{sample} \text{ fixed point up to } o(|r_t|)\text{).}$$

Interpretation: C measures local sensitivity. As $C \uparrow 1$, δ_{iter} inflates and the fixed point becomes poorly conditioned.

Proposition (A Posteriori bound)

If T is C^1 and is a contraction on a neighborhood $\mathcal{N}$ of $\widehat{\mathcal{Z}}$ with Lipschitz $L < 1$, then for any $z \in \mathcal{N}$,

$$|z - \widehat{\mathcal{Z}}| \leq \frac{|T(z) - z|}{1 - L}.$$

At termination ($z = z^{(t)}$) and with $C = |T'(\widehat{\mathcal{Z}})|$,

$$|z^{(t)} - \widehat{\mathcal{Z}}| \leq \frac{|r_t|}{1 - C} + o(|r_t|).$$

A Posteriori Bounds

Noise Amplification (Sensitivity of the fixed point)

For a smooth perturbation $T_\lambda(z) = T(z) + \lambda\,\Delta(z)$,

$$\left.\frac{d\widehat{\mathcal{Z}}_\lambda}{d\lambda}\right|_{\lambda=0} = \frac{\Delta(\widehat{\mathcal{Z}})}{1 - T'(\widehat{\mathcal{Z}})}.$$

Takeaway: map noise is geometrically amplified by the conditioning factor $1/(1-C)$.

Tolerance To Error

Assume stopping rule $|z^{(t+1)} - z^{(t)}| \le \varepsilon\, z^{(t)}$ and local contractivity.

Tolerance-to-error guarantee

$$\frac{|z^{(t)} - \widehat{Z}|}{z^{(t)}} \le \frac{\varepsilon}{1-C} + o(\varepsilon).$$

Using the diagnostic alongside MC uncertainty

Define $U_{\mathrm{iter}} := \delta_{\mathrm{iter}}/\sqrt{\mathrm{Var}_{\mathrm{MC}}}$.

- If $U_{\mathrm{iter}} > 1$: iteration error can *dominate* the Monte Carlo standard error.
- If $C \ge 0.9$: ill-conditioned; $1/(1-C) \ge 10$ implies order-of-magnitude amplification of map noise.
- "Safe" example: $C \le 0.5$ and $U_{\mathrm{iter}} \le 0.25 \Rightarrow$ iteration error $\le 25\%$ of MCSE.

Table of Contents

Delta Through The Fixed Point

Let $F(z,\vartheta) := z - T(z \mid \vartheta)$ and suppose $\widehat{\vartheta}$ is fitted.

$$\frac{\partial \widehat{\mathcal{Z}}}{\partial \vartheta} = \frac{\partial_\vartheta T(\widehat{\mathcal{Z}} \mid \widehat{\vartheta})}{1 - \partial_z T(\widehat{\mathcal{Z}} \mid \widehat{\vartheta})} = \frac{\partial_\vartheta T(\widehat{\mathcal{Z}} \mid \widehat{\vartheta})}{1 - T'(\widehat{\mathcal{Z}})}. \tag{9}$$

If $T = A/B$ with $A(z) = n(z)$ and $B(z) = d(z)$,

$$\partial_\vartheta T(\widehat{\mathcal{Z}} \mid \widehat{\vartheta}) = \left.\frac{(\partial_\vartheta A)B - A(\partial_\vartheta B)}{B^2}\right|_{z=\widehat{\mathcal{Z}}}. \tag{10}$$

Using $\partial_\vartheta \ell = -\ell\, \partial_\vartheta \log g$,

$$\partial_\vartheta A(z) = \frac{s_2 z}{S_2} \sum_{i=1}^{S_2} \frac{\partial_\vartheta \ell_{2,i}}{\left(s_1 \ell_{2,i} + s_2 z\right)^2}, \qquad \partial_\vartheta B(z) = -\frac{s_1}{S_1} \sum_{j=1}^{S_1} \frac{\partial_\vartheta \ell_{1,j}}{\left(s_1 \ell_{1,j} + s_2 z\right)^2}. \tag{11}$$

Assume a regular estimator $\widehat{\vartheta}$ with $\sqrt{\mathrm{neff}}(\widehat{\vartheta} - \vartheta_0) \Rightarrow \mathcal{N}(0, \Sigma_\vartheta)$, where neff is the number of effective draws.

Asymptotic Variance Decomposition

$$\mathrm{Var}(\widehat{\mathcal{Z}}) = \underbrace{\mathrm{Var}_\delta(\widehat{\mathcal{Z}})}_{\text{conditional MC (delta)}} + \underbrace{\left(\partial_\vartheta \widehat{\mathcal{Z}}\right) \frac{\Sigma_\vartheta}{\mathrm{neff}} \left(\partial_\vartheta \widehat{\mathcal{Z}}\right)^\top}_{\mathrm{Var}_g(\widehat{\mathcal{Z}})} + o\big(S_1^{-1} + S_2^{-1} + \mathrm{neff}^{-1}\big).$$

Then one can define the inflation ratio

$$R_g := \frac{\sqrt{\mathrm{Var}_g(\widehat{\mathcal{Z}})}}{\sqrt{\mathrm{Var}_\delta(\widehat{\mathcal{Z}})}} \Rightarrow \frac{\mathrm{MCSE}_{\text{total}}}{\sqrt{\mathrm{Var}_\delta(\widehat{\mathcal{Z}})}} \approx \sqrt{1 + R_g^2}. \tag{12}$$

Table of Contents

Conclusions

- Three **drop-in** diagnostics for bridge sampling that quantify:
  1. Influence/leverage and unconditional variance ($\mathcal{V}_{\text{jack}}$, MaxInfluence).
  2. Ill-conditioning and iteration error (C, U_{iter}).
  3. Added variance from proposal fitting (R_g).
- Each has **transparent thresholds** with finite-sample control or asymptotic calibration.
- They **complement** (not replace) the classical delta MCSE and standard overlap checks.

Limitations And Open Directions

- Jackknife assumes block sizes relies on mild mixing; heavy-tailed posteriors may require robustified blocks.
- Conditioning factor C is local; global non-contractive behavior needs additional safeguards.
- Proposal-fit diagnostic depends on a covariance estimate for $\widehat{\vartheta}$; poor fit models can understate Σ_{ϑ}.
- **Open**: automated proposal enrichment under diagnostic feedback, non-asymptotic finite-sample bounds tailored to u_2, v_1 ranges.